COLUMBIA UNIVERSITY
IN THE CITY OF NEW YORK

STAT 5703

Statistical Inference and Modeling for Data Science

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Example (From last time):

Let $X_1, X_2, \ldots, X_n$ be a random sample from the $U(0, \theta)$ distribution, where θ is unknown. Define the estimator

$$\hat{\theta} = \max\{X_1, \ldots, X_n\}$$

Answers to Example 5:

a) $\text{Bias}(\hat{\theta}) = -\frac{\theta}{n+1}$

b) $\text{Var}(\hat{\theta}) = \frac{n\theta^2}{(n+2)(n+1)^2}$

Example 1:

Let $X_1, X_2, \ldots, X_n$ be a random sample from the $U(0, \theta)$ distribution, where θ is unknown. Define the estimator

$$\tilde{\theta} = 2\,\bar{X}$$

Find:

a) $\text{Bias}(\tilde{\theta}) =$

b) $\text{Var}(\tilde{\theta}) =$

Recall:

If $Y \sim \text{U}(a, b)$ then $E(Y) = \frac{a+b}{2}, \text{Var}(Y) = \frac{(b-a)^2}{12}$

Example 1 (Solution):

Let $X_1, X_2, \ldots, X_n$ be a random sample from the $U(0, \theta)$ distribution, where θ is unknown. Define the estimator

$$\tilde{\theta} = 2\,\bar{X}$$

Find:

a) $\text{Bias}(\tilde{\theta}) = E(\tilde{\theta}) - \theta = 2\text{E}(\bar{X}) - \theta$

$$= 2\mu - \theta = 2\frac{0+\theta}{2} - \theta = \theta - \theta = 0$$

b) $\text{Var}(\tilde{\theta}) = \text{Var}(2\bar{X}) = 4\text{Var}(\bar{X})$

$$= 4\frac{\sigma^2}{n} = 4\frac{(\theta - 0)^2}{12n} = \frac{\theta^2}{3n}$$

How are the formulas for these estimators derived?

The most common methods are :

- The method of moments (MM)
- Maximum likelihood estimators (MLE)
- Minimizing sum of squared errors (LS)

Moments of a Distribution

Definition: Let $X_1, X_2, \ldots, X_n$ be a random sample from the population random variable X with pdf $f(x; \theta)$. The kth *population* moment is

$$\mu_k = E(X^k)$$

Similarly, the kth *sample* moment is

$$\hat{\mu}_k = \frac{1}{n}\sum_{i=1}^{n} X_i^k, k = 1, 2, \ldots$$

Note: These moments depend on the unknown parameters, so $\hat{\mu}_k = \mu_k(\theta_1, \ldots, \theta_m)$.

Method of Moments

Algorithm: Suppose that we have m unknown parameters to be estimated. That is, $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_m)$. The method of moments estimators $\hat{\theta}_1, \ldots, \hat{\theta}_m$ are obtained by solving for $\theta_1, \ldots, \theta_m$ the system of equations

$$\mu_1(\theta_1, \ldots, \theta_m) = \hat{\mu}_1$$
$$\vdots$$
$$\mu_m(\theta_1, \ldots, \theta_m) = \hat{\mu}_m$$

Example 2:

Let $x_1 = 0.4$, $x_2 = 0.7$, $x_3 = 0.9$ be a sample from the pdf

$$f(x; \theta) = \theta x^{\theta - 1},\ 0 < x < 1$$

Estimate θ by the method of moments.

$$\Rightarrow \bar{x} = \frac{0.4 + 0.7 + 0.9}{3} = \frac{2}{3}$$

Solution:

In this case we have one unknown parameter, so $m = 1$

$$\mu_1 = E(X) = \int_0^1 x f(x;\theta)\,dx = \int_0^1 x\theta x^{\theta-1}\,dx$$

$$= \theta \int_0^1 x^{\theta}\,dx = \theta \left.\frac{x^{\theta+1}}{\theta+1}\right|_0^1 = \frac{\theta}{\theta+1}$$

$\Rightarrow$

$$\mu_1(\theta) = \frac{\theta}{\theta+1} \approx \bar{X} = \hat{\mu}_1$$

$\Rightarrow$ Solve for θ: $\frac{\theta}{\theta+1} = \bar{x} \Leftrightarrow \theta = \theta\bar{x} + \bar{x}$

$\theta(1-\bar{x}) = \bar{x}$

$$\Rightarrow \boxed{\hat{\theta}_{MM} = \frac{\bar{x}}{1-\bar{x}}} = \frac{2/3}{1-2/3} = \boxed{2}$$

Exercise 1:

Use the method of moments to estimate θ in the pdf

$$f(x; \theta) = (\theta^2 + \theta)x^{\theta - 1}(1 - x), \quad 0 < x < 1$$

Consistency of MM

Note that at the end, if the MM estimator exists and if it is unique, it has the form

$$\hat{\theta}_{MM} = h(\bar{X})$$

Therefore, by the continuous mapping theorem, if h is a continuous function, we have that:

$$\hat{\theta}_{MM} \xrightarrow{P} h(\mu) = \theta$$

This argument can be extended to more than one parameter.

Example 3

Let $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$ with density function

$$f(x;\lambda) = \lambda e^{-\lambda x}, x \geq 0, \lambda > 0$$

Find the MM estimator of λ.

Solution: If $X \sim \text{Exp}(\lambda)$, then $E(X) = \frac{1}{\lambda}$, therefore

$$\hat{\lambda} = \frac{1}{\bar{X}}$$

Example 3 (discussion):

Let $X_1, \ldots, X_n \sim \text{Exp}(\lambda)$. When using the first moment, the MM estimator of λ is:

$$\hat{\lambda} = \frac{1}{\bar{X}}$$

What if we use the second moment?

$$E(X^2) = \frac{2}{\lambda^2} = \hat{\mu}_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 \Rightarrow \tilde{\lambda} = \sqrt{\frac{2}{\hat{\mu}_2}}$$

Note: Theoretical arguments connected to efficiency of estimators justify starting from lower order moments.

Example 4

Let $X_1, \ldots, X_n$ be a random sample from X which follows the Pareto model:

$$f(x;\theta) = \frac{\theta}{x^{\theta+1}}, x > 1, \theta > 0$$

Show as exercise that

$$\mu_1 = E(X) = \frac{\theta}{\theta-1} \;\&\; Var(X) = \frac{\theta}{(\theta-1)^2(\theta-2)}$$

Find the MM estimator of θ and its asymptotic distribution.

Example 4 (Solution):

$$\mu_1 = E(X) = \frac{\theta}{\theta - 1} = \bar{X} \Leftrightarrow \theta = \theta\bar{X} - \bar{X}$$

$$\Rightarrow \ \hat{\theta}_{MM} = h(\bar{X}) = \frac{\bar{X}}{\bar{X} - 1}$$

We know by CLT that $\sqrt{n}\left(\bar{X} - \frac{\theta}{\theta-1}\right) \xrightarrow{d} N\left(0, \sqrt{\frac{\theta}{(\theta-1)^2(\theta-2)}}\right)$.

Then by the delta method

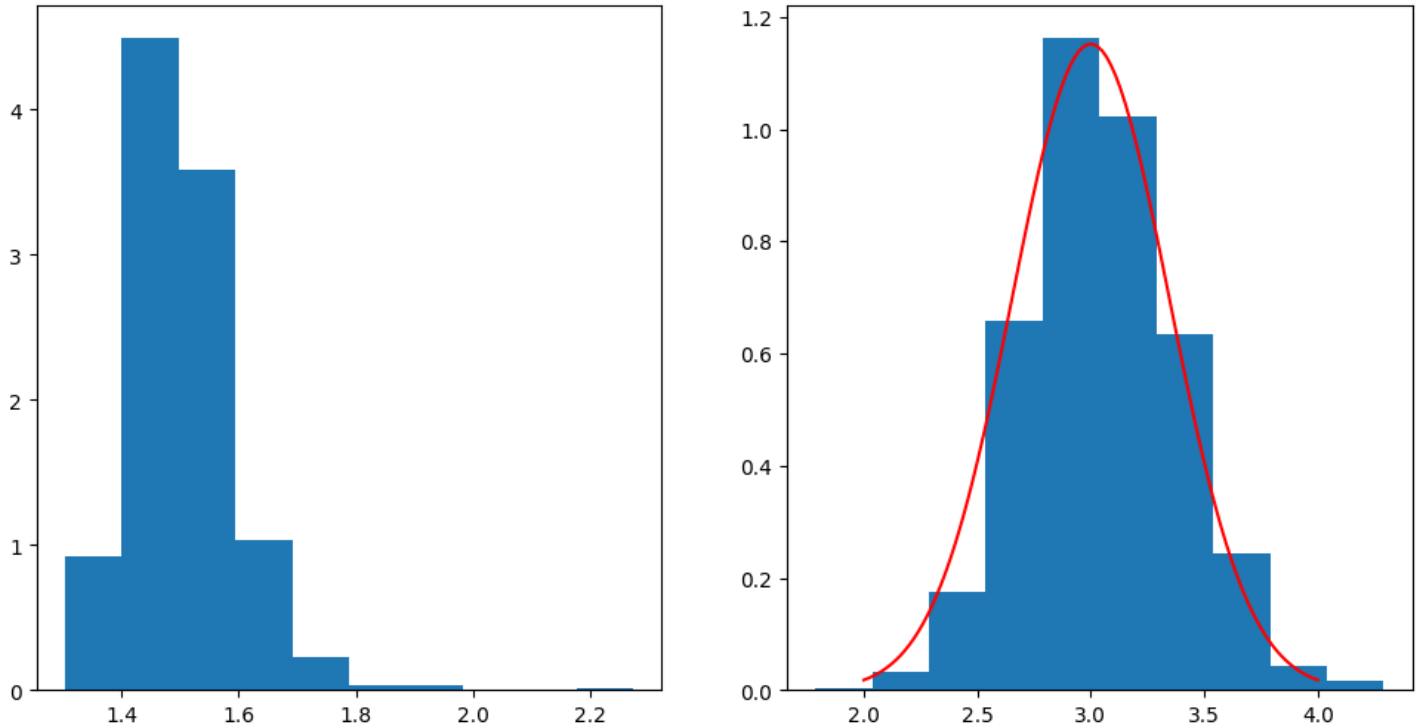
$$\sqrt{n}\left(h(\bar{X}) - h\left(\frac{\theta}{\theta - 1}\right)\right) \xrightarrow{d} N\left(0, \sqrt{\frac{\theta}{(\theta-1)^2(\theta-2)}\left[h'\left(\frac{\theta}{\theta-1}\right)\right]^2}\right)$$

$$\Leftrightarrow \sqrt{n}\left(\hat{\theta}_{MM} - \theta\right) \xrightarrow{d} N\left(0, \sqrt{\frac{\theta}{(\theta-1)^2(\theta-2)}\left[-\frac{1}{\left(\frac{\theta}{\theta-1} - 1\right)^2}\right]^2}\right)$$

$$\Leftrightarrow \sqrt{n}\left(\hat{\theta}_{MM} - \theta\right) \xrightarrow{d} N\left(0, \sqrt{\frac{\theta(\theta-1)^2}{(\theta-2)}}\right)$$

Example 4 (Solution):

Verify the result with Python simulation



- Left panel: simulated distribution of $\bar{X}$ from Pareto(3) with n = 100
- Right panel: simulated distribution of $\hat{\theta}_{MM}$
- Overlayed curve is $N\left(0, \sqrt{\frac{\theta(\theta-1)^2}{n\,(\theta-2)}}\right)$

Example 5

Suppose we have sample from the Gamma distribution:

$$X_1, \dots, X_n \sim \Gamma(\alpha, \beta)$$

Find the MM estimators of α, β

Hint:

$$\mu_1 = E(X) = \frac{\alpha}{\beta} \;\&\; Var(X) = \mu_2 - \mu_1^2 = \frac{\alpha}{\beta^2}$$

Example 5 (Solution)

Let $X_1, \ldots, X_n \sim \Gamma(\alpha, \beta)$.

Find the MM estimators of α, β

Recall: $\mu_1 = E(X) = \frac{\alpha}{\beta}$ & $Var(X) = \mu_2 - \mu_1^2 = \frac{\alpha}{\beta^2}$

Step 1: Express α from the first equation as a function of β and μ_1: $\alpha = \beta\mu_1$

Step 2: Substitute α in the second equation:

$$\frac{\beta\mu_1}{\beta^2} = \mu_2 - \mu_1^2 \Rightarrow \beta = \frac{\mu_1}{\mu_2 - \mu_1^2}$$

Step 3: Replace μ_1 and μ_2 with their sample estimators:

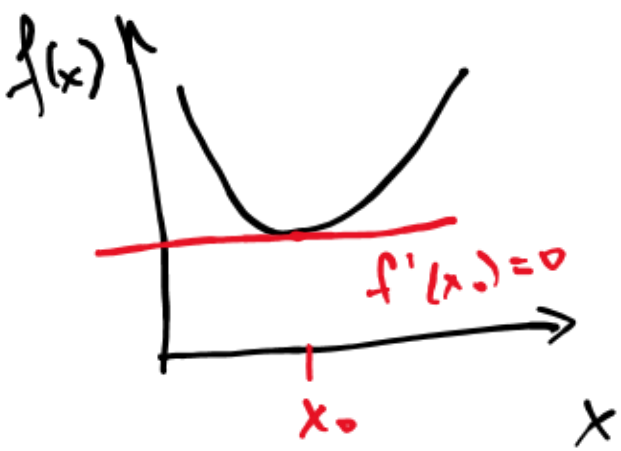
$$\hat{\beta}_{MM} = \frac{\bar{X}}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2} \text{ \& } \hat{\alpha}_{MM} = \frac{\bar{X}^2}{\frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2}$$

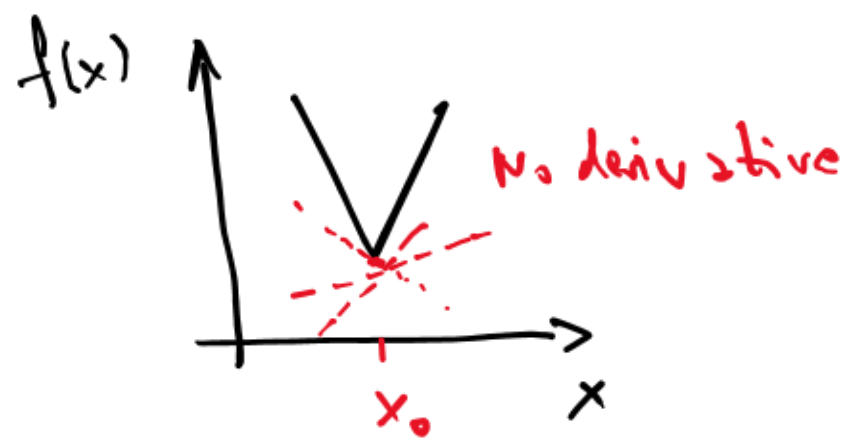
Review of
Maximum Likelihood Estimation
from STAT 5701

Recall: Extrema of Functions

Theorem 1 (Fermat): If x_0 is a local *extremum point* of $f(x)$ and $f(x)$ is differentiable at x_0, then $f'(x_0) = 0$.

Note: $f'(x_0)$ might not exist and we can still have a local extremum at x_0.





Second Derivative Test

Theorem 2:

Let $f'(x_0) = 0$ and $f''(x_0)$ exists. Then

if $f''(x_0) > 0$ then x_0 is a min of $f(x)$

if $f''(x_0) < 0$ then x_0 is a max of $f(x)$

Remark: If $f''(x_0) = 0$, then you have to find $f'''(x_0)$ and if $f'''(x_0) \neq 0$, then x_0 is an inflection point.

Inflection point:

$f(x) = x^3$

Exercise 2:

Find max $f(\theta) = \frac{\theta}{\theta^2+1}$ in $\theta \in (0,2)$

Maximum Likelihood Estimation

Definition 1: Likelihood function of θ is the *joint* pdf of $X_1, \ldots, X_n$:

$$L(\theta) = f(x_1, \ldots, x_n; \theta) = \prod_{i=1}^{n} f(x_i; \theta)$$

since X_i's are independent.

Note: $L(\theta)$ is aka *evidence*.

Basic idea: Estimate θ with $\hat{\theta}$ such that $L(\theta)$ is maximized at $\hat{\theta}$ for the observed data.

Maximum Likelihood Estimation

Definition 2: MLE of θ is

$$\hat{\theta}_{MLE} = \arg\max L(\theta)$$

That is, θ which maximizes globally the likelihood $L(\theta)$.

Log-Likelihood and Score Function

In general, it is easier to find $\hat{\theta}$ from the *log-likelihood*, defined as

$$l(\theta) = \log\big(L(\theta)\big)$$

wherever $L(\theta) \neq 0$.

It can be shown that $L(\theta)$ and $l(\theta)$ achieve their maximum at the same point $\hat{\theta}$.

The *score* function, needed to derive the MLE, is defined as

$$s(\theta) = \frac{\partial l(\theta)}{\partial \theta}$$

Most often we find the MLE $\hat{\theta}$ as the solution to $s(\theta) = 0$

Vector Parameter Space

Suppose we have a vector of parameters $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_m)$. In this case the score function will be an $m \times 1$ gradient vector, and the second derivatives will be an $m \times m$ matrix.

Definition: Let $X_1, \ldots, X_n$ be a sample with log-likelihood $l(\boldsymbol{\theta})$. The *score vector* is defined as

$$s(\boldsymbol{\theta}) = \nabla_{\boldsymbol{\theta}} l(\boldsymbol{\theta}) = \left(\frac{\partial}{\partial \theta_1} l(\boldsymbol{\theta}), \ldots, \frac{\partial}{\partial \theta_m} l(\boldsymbol{\theta}) \right)^T$$

and the *Hessian matrix* is defined as

$$H(\boldsymbol{\theta}) = \left\{ \frac{\partial^2}{\partial \theta_i \partial \theta_j} l(\boldsymbol{\theta}) \right\}$$

Likelihood for Censored Data

Suppose that a random variable $X \sim N(\mu, \sigma)$, but whenever $X \leq 0$, all that we observe is that it is less than or equal to 0. We say that X has been *censored* at 0. This is a very common scenario in survival analysis.

Assume we have a sample of size n of which n_0 observations are censored at 0 and the rest fully observed.

Q: What is the likelihood of the observed data

A:

$$L(\mu, \sigma) = \Phi\left(-\frac{\mu}{\sigma}\right) \prod_{i=1}^{n-n_0} \left[\frac{1}{\sigma}\phi\left(\frac{x_i - \mu}{\sigma}\right)\right]$$

Where Φ is the cdf of a standard normal variable, and ϕ is the pdf of a standard normal variable.

Example 3 Exponential, continued (from last semester):

$$X_1, X_2, \dots, X_n \sim Exp(\lambda)$$

Assume we have the following data:

225, 171, 198, 189, 189, 135, 162, 135, 117, 162

a) Derive and graph the likelihood function

b) Find the MLE of λ.

Example 3 (solution – read on your own):

$$X_1, X_2, \dots, X_n \sim Exp(\lambda)$$

$$\Rightarrow L(\lambda) = \prod_{i=1}^{n} \lambda e^{-\lambda x_i} = \left(\prod_{i=1}^{n} \lambda\right)\left(\prod_{i=1}^{n} e^{-\lambda x_i}\right)$$

$$= \lambda^n e^{\sum_{i=1}^{n}(-\lambda x_i)} = \lambda^n e^{-\lambda \sum_{i=1}^{n} x_i}$$

$$\Rightarrow l(\lambda) = \log \lambda^n + \log e^{-\lambda \sum_{i=1}^{n} x_i} = n \log \lambda - \lambda \sum_{i=1}^{n} x_i$$

$$\Rightarrow s(\lambda) = l'(\lambda) = n\frac{1}{\lambda} - \sum_{i=1}^{n} x_i \Rightarrow \hat{\lambda}_{MLE} = \frac{n}{\sum x_i} = \frac{1}{\bar{x}} = \frac{1}{168.3}$$

Is it a local max?

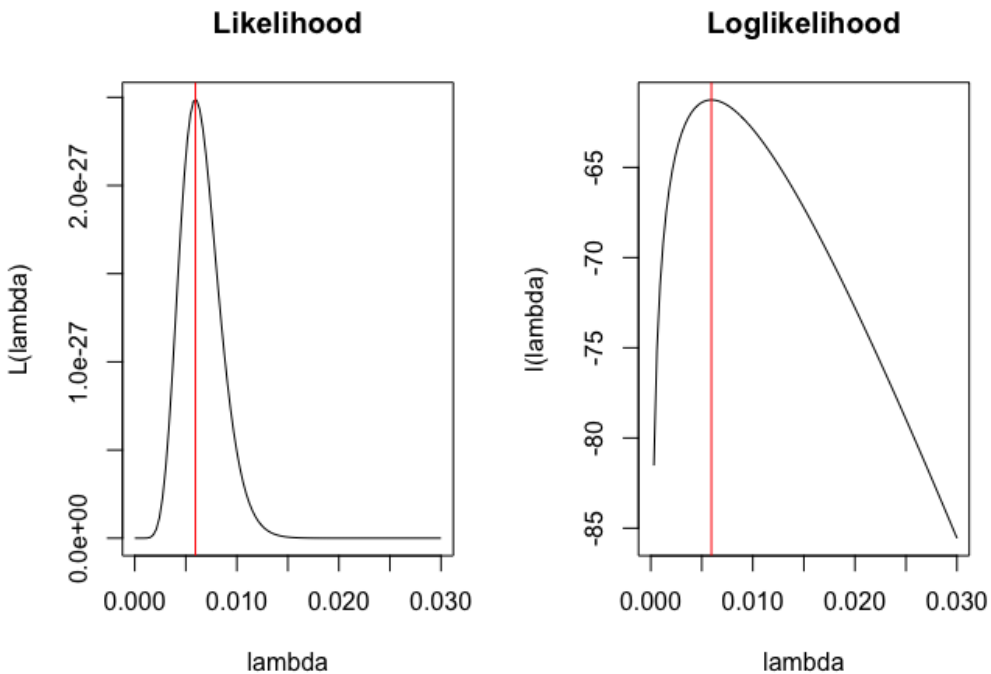
$$l''(\lambda) = -n\frac{1}{\lambda^2} < 0, \forall \lambda$$

Example 3 (solution):

Assume we have the following data:

225, 171, 198, 189, 189, 135, 162, 135, 117, 162

a) graph the likelihood function



Exercise 3 (Poisson):

$$X_1, X_2, \dots, X_n \sim Poisson(\lambda)$$

Find the MLE of λ.

$$P(x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

Example 6: Logistic Regression

- Assume $Y_i|x_i \sim Bernoulli\big(p(x_i)\big)$, where the model is:

$$\log\frac{p(x_i)}{1-p(x_i)} = \beta_0 + \beta_1 x_i$$

- Then the likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p(x_i)^{y_i}[1-p(x_i)]^{1-y_i}$$

Since

$$\log\frac{p(x_i)}{1-p(x_i)} = \beta_0 + \beta_1 x_i$$

$$\Rightarrow \frac{p(x_i)}{1-p(x_i)} = e^{\beta_0+\beta_1 x_i} \Rightarrow p(x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1+e^{\beta_0+\beta_1 x_i}}$$

$$\Rightarrow 1-p(x_i) = \frac{1}{1+e^{\beta_0+\beta_1 x_i}}$$

Logistic Regression: Estimating Betas

- When $Y_i|x_i \sim Bernoulli\big(p(x_i)\big)$, then the likelihood function is:

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} p(x_i)^{y_i}[1 - p(x_i)]^{1-y_i}$$

$$p(x_i) = \frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}}, 1 - p(x_i) = \frac{1}{1 + e^{\beta_0+\beta_1 x_i}}$$

and therefore

$$L(\beta_0, \beta_1) = \prod_{i=1}^{n} \left(\frac{e^{\beta_0+\beta_1 x_i}}{1 + e^{\beta_0+\beta_1 x_i}}\right)^{y_i} \left(\frac{1}{1 + e^{\beta_0+\beta_1 x_i}}\right)^{1-y_i}$$

$$= \prod_{i=1}^{n} \left[\left(e^{\beta_0+\beta_1 x_i}\right)^{y_i} \frac{1}{1 + e^{\beta_0+\beta_1 x_i}}\right]$$

$$\Rightarrow l(\beta_0, \beta_1) = \sum_{i=1}^{n} y_i(\beta_0 + \beta_1 x_i) - \sum_{i=1}^{n} \log\left(1 + e^{\beta_0+\beta_1 x_i}\right)$$

Logistic Regression: Estimating Betas

- The logistic regression loglikelihood is:

$$l(\beta_0, \beta_1) = \sum_{i=1}^{n} y_i(\beta_0 + \beta_1 x_i) - \sum_{i=1}^{n} \log\left(1 + e^{\beta_0 + \beta_1 x_i}\right)$$

$$\Rightarrow \frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) = 0$$

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} \left(y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \left(y_i - p(x_i) \right) x_i = 0$$

- Unlike the closed-form analytical solution (i.e., the normal equations) available for linear regression, the score function for logistic regression is transcendental. That is, there is no closed-form solution so we must solve with numerical optimization methods such as Newton's method (IRLS) which is done in several iterations.
- Although we have focused on simple logistic regression, this likelihood (and score) function generalize directly to the case of *p* predictors.

Logistic Regression: Estimating Betas

- The simple logistic regression estimating equations are:

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_0} = \sum_{i=1}^{n} \left(y_i - \frac{e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \big(y_i - p(x_i) \big) = 0$$

$$\frac{\partial l(\beta_0, \beta_1)}{\partial \beta_1} = \sum_{i=1}^{n} \left(y_i x_i - \frac{x_i e^{\beta_0 + \beta_1 x_i}}{1 + e^{\beta_0 + \beta_1 x_i}} \right) = \sum_{i=1}^{n} \big(y_i - p(x_i) \big) x_i = 0$$

- In general, the gradient is:

$$\frac{\partial l(\boldsymbol{\beta})}{\partial \beta_j} = \sum_{i=1}^{n} \big(y_i - p(x_i) \big) x_{ij}$$

- Gradient Ascent Optimization:

$$\beta_j^{new} = \beta_j^{old} + \eta \frac{\partial l\big(\boldsymbol{\beta}^{old}\big)}{\partial \beta_j^{old}}$$

where η is the magnitude of the step size that we take.